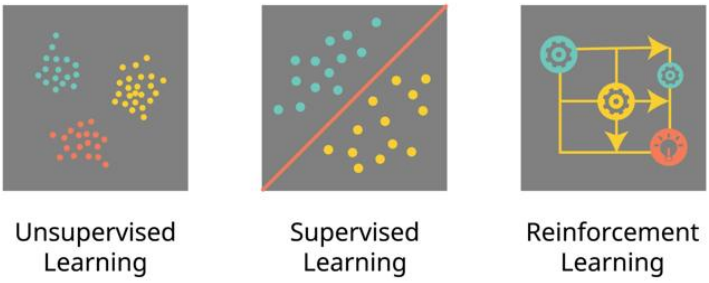
Linear Regression

1. What are the main parts of Machine Learning?



2. What is supervised learning algorithm?

A supervised learning algorithm is a computer program that learns to make predictions by studying historical examples that already include the correct answers.

3. What are the main types of supervised learning?

Classification: Used when the answer is a category or a group.

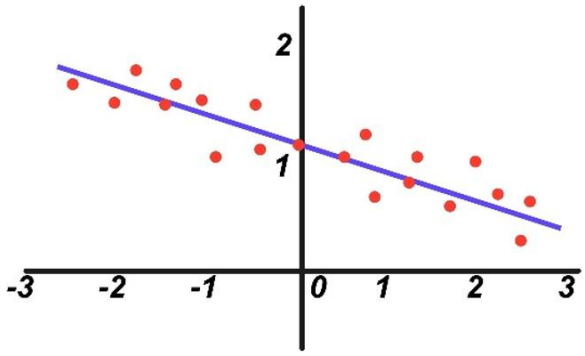
- Example: Looking at an incoming message and predicting if it is “Spam” or “Not Spam”.

Regression: Used when the answer is a continuous number.

- Example: Looking at a house’s square footage to predict its specific dollar price.

4. What is Linear Regression?

Linear regression is a machine learning algorithm that predicts a specific numerical value by finding the straight “line of best fit” that connects the input data to its output.



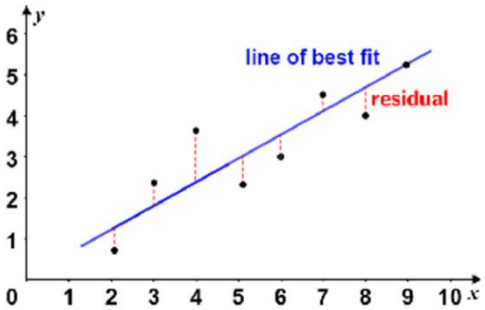
$y = wx + b$

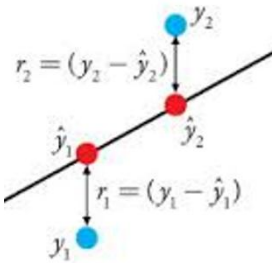
- y: output (dependent variable)
- x: input (independent variable)
- w: slope (weight)
- b: y-intercept (bias)

5. How to obtain the "line of best fit"?

If there is a scattered cluster of dots on a graph, we can draw thousands of different straight lines through them. So, this is the way to find the best line;

Residual error: When a line is drawn through the data points, most of the dots won't land perfectly on the line. The vertical distance between the actual dot(real data) and the line(predicted data) is called the residual.





$$\text{residual} = y_{\text{actual}} - y_{\text{predicted}}$$

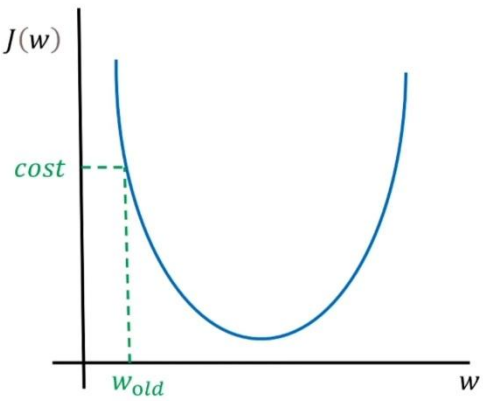
If y(actual) is bigger than y(predicted), then residual is a positive value.
If y(actual) is smaller than y(predicted), then residual is a negative value.

To stop the positive and negative errors from cancelling each other out, every single residual is squared before adding them together. After that the total value is divided by the number of data points. It's the cost function (error)

$$\text{cost function(error)} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2$$

The algorithm tests different values for the slope(w) and bias(b) and calculates the cost function. The specific line that produces the lowest cost function is officially considered as the "line of best fit".

Gradient Descent Algorithm

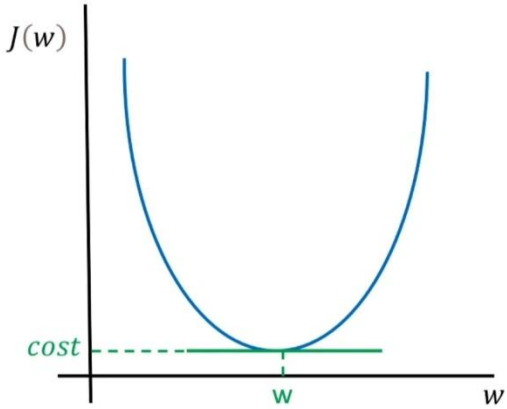


$$w_{new} = w_{old} - step\ size$$

$$w_{new} = w_{old} - learning\ rate\ *\ slope$$

We keep the learning rate (alpha) small to ensure the algorithm makes steady, safe progress down the curve without overshooting the lowest point. If the value is too large, the algorithm

takes massive steps and can jump right over the bottom, bouncing back and forth on the sides forever. Keeping it small guarantees, we cautiously step downward until we accurately settle at the minimum cost.



$$w_{new} = w_{old} - step\ size$$

$$w_{new} = w_{old} - learning\ rate\ *\ slope$$

$$w_{new} = w_{old} - 0$$

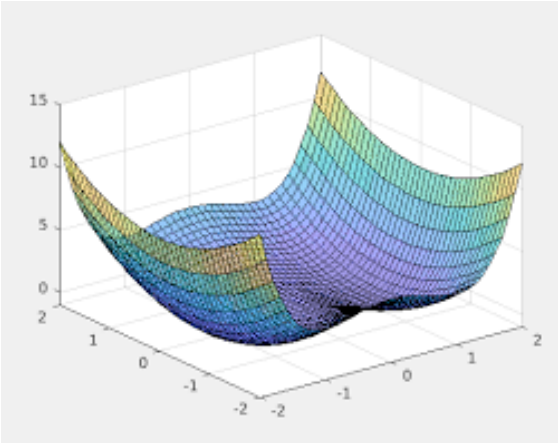
When the algorithm reaches the exact bottom of the U-shaped curve, it has found the optimal solution where the model's errors are minimized. At this lowest point, the curve is perfectly flat, meaning the slope (gradient) is exactly zero. Because the slope is zero, the equation calculates a step size of zero (w(new) = w(old) - 0), causing the parameter w to stop updating entirely. This is called "convergence" and it signals that the algorithm is finished because it has successfully locked in the best possible, fully optimized weights.

Similar procedure is followed to calculate the bias term also.

$$b_{new} = b_{old} - learning\ rate\ *\ slope$$

6. How to handle Multiple Linear Regression?

$$y = w_1x_1 + w_2x_2 + w_3x_3 + \cdots + w_nx_n + b$$



In this 3D space, the height of the bowl represents the Cost (how wrong the model is). The flat floor represents our two weights (w(1) and w(2)). The algorithm starts high up on the edge of the bowl and takes calculated steps down the steepest slope until it reaches the lowest point at the very bottom. That bottom point gives us the optimized weights for the best-fitting 3D plane.

7. What is Regularization?

Regularization is a mathematical technique used to control a machine learning model by adding a “penalty” to its cost function. Instead of just letting the algorithm do whatever it takes to minimize errors, Regularization forces the model to keep its weights (w) as small and simple as possible.

8. Why do we want Regularization?

To prevent overfitting: When a model is trained with many variables, it can try too hard to perfectly connect every single dot in the training data. This results in a wild, squiggly line that has memorized the training data (including the random “noise” and outliers) but completely fails to predict new, real-world data.

9. What are the main methods of Regularization?

Method	Penalty Type	How it works	The Result	When to use
Ridge	L2 Penalty (Square of weights)	Shrinks all weights toward zero, but never completely to zero.	Keeps all variables in the equation but reduces their impact.	When you believe most of your input variables are useful and you just want to prevent extreme weights.
Lasso	L1 Penalty (Absolute value)	Shrinks weights aggressively, pushing some all the way to absolute zero.	Deletes useless variables entirely, acting as a built-in "cleanup" tool.	When you have a massive dataset and suspect many of the inputs are garbage or irrelevant.
ElasticNet	L1 + L2 (Combination)	Uses a mixing ratio to apply both penalties at the same time.	Safely shrinks correlated variables together while still deleting the totally useless ones.	When you have a complex dataset with many inputs that might be highly correlated with each other.